Regulatory systems Maritime Innovation

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Definitions

- “innovation, the creation of a new way of doing something, whether the enterprise is concrete (e.g., the development of a new product) or abstract (e.g., the development of a new philosophy or theoretical approach to a problem).”
  - Encyclopedia Britannica

- “regulation, in government, a rule or mechanism that limits, steers, or otherwise controls social behaviour.”
  - Encyclopedia Britannica

The questions to be discussed

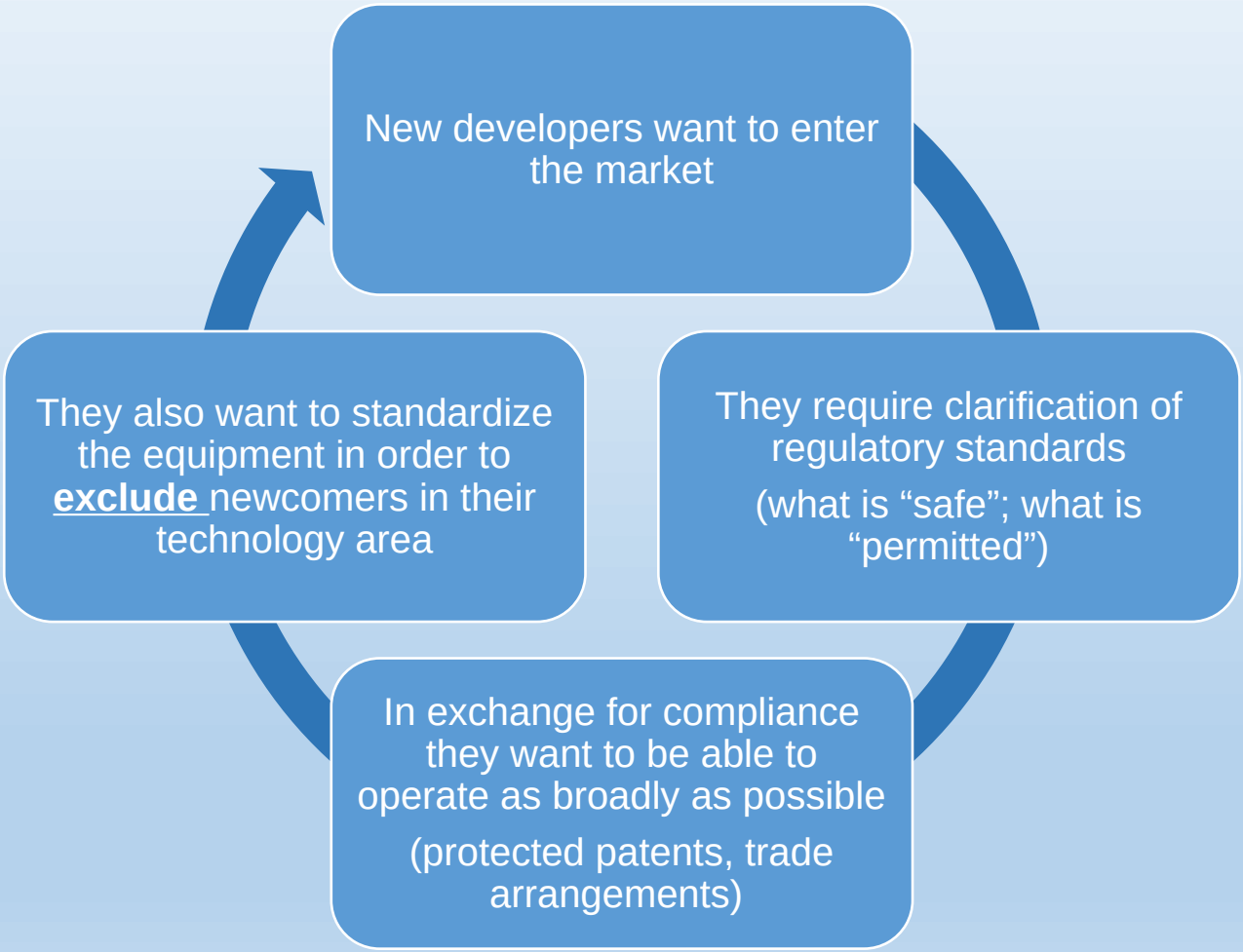
- How do regulation and innovation interact?
- Who makes regulations and how do they relate to technology, ethics and social beliefs?
- Are there regulatory approaches which support innovation?
- Should society adopt such approaches?

The Locomotive Act, 1865

- s. 3 Rules for the manner of working locomotives on turnpike roads and turnpikes herein stated. Every locomotive propelled by steam or any other than animal power on any turnpike road or public highway shall be worked according to the following rules and regulations, viz.:–
- (1) At least three persons shall be employed to drive or conduct such locomotive, and if more than two waggons or carriages be attached, thereto, an additional person shall be employed, who shall take charge of such waggons or carriages.
- (2) One of such persons, while any locomotive is in motion, shall precede such locomotive on foot by not less than sixty yards, and shall carry a red flag constantly displayed, and shall warn the riders and drivers of horses of the approach of such locomotives, and shall signal the driver thereof when it shall be necessary to stop, and shall assist horses, and carriages drawn by horses, passing the same.
- (3) The drivers of such locomotives shall give as much space as possible for the passing of other traffic.
- (4) The whistle of such locomotive shall not be sounded for any purpose whatever; nor shall the cylinder taps be opened within sight of any person riding, driving, leading or in charge of a horse upon the road; nor shall the steam be allowed to attain a pressure such as to exceed the limit fixed by the safety valve, so that no steam shall blow off when the locomotive is upon the road.
- (5) Every such locomotive shall be instantly stopped, on the person preceding the same, or any other person with a horse, or carriage drawn by a horse, putting his hand as a signal to require such locomotive to be stopped.
- (6) Any person in charge of any such locomotive shall provide two efficient lights to be affixed conspicuously, one at each side on the front of the same, between the hours of one hour after sunset and one hour before sunrise.
- In the event of a non-compliance with any of the provisions of this section, the owner of the locomotive shall, on summary conviction thereof before to justices, be liable to pay a penalty not exceeding £10; but it shall be lawful for such owner, on proving that he has incurred such penalty by reason of the negligence or wilful default of any person in charge of or in attendance on such locomotive, to recover summarily from such person the whole or any part of the penalty he may have incurred as owner.
- 4 Limit of speed of locomotives on turnpike roads and highways.
- Subject and without prejudice to the regulations hereinafter authorised to be made by local authorities, it shall not be lawful to drive any such locomotive along any turnpike road or public highway at a greater speed than four miles an hour, or through any city, town, or village at a greater speed than two miles an hour; and any person acting contrary thereto shall for every such offence, on summary conviction thereof, forfeit any sum not exceeding £10.

Why was this law made?
Who benefitted from it?

Regulatory pressures- (1) New Technology Companies



Regulatory pressures- (2) Conventional Technology

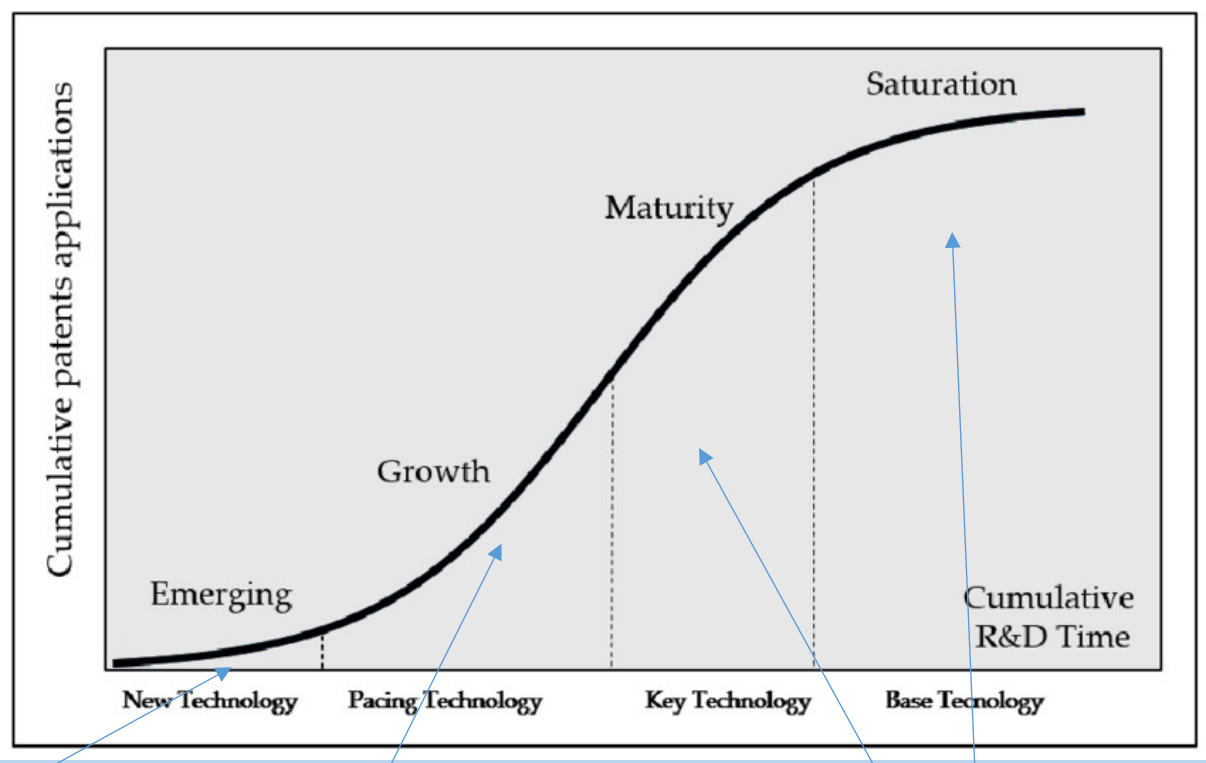
Existing traders wish to protect their investments and market share

They support existing regulatory standards to avoid costs and state safety issues

(the “unknown”)

They insist on standardization for maximum trading scope

The innovation “cycle”



Unregulated and general law

Quick Regulatory evolution. Support for this innovation

Established Regulatory system
Resistance to competitive innovation

Examples

What are the ethical, moral and legal issues in each example?
How large is each market?

Maritime innovation and Regulation
Autonomous ships

The argument for autonomy

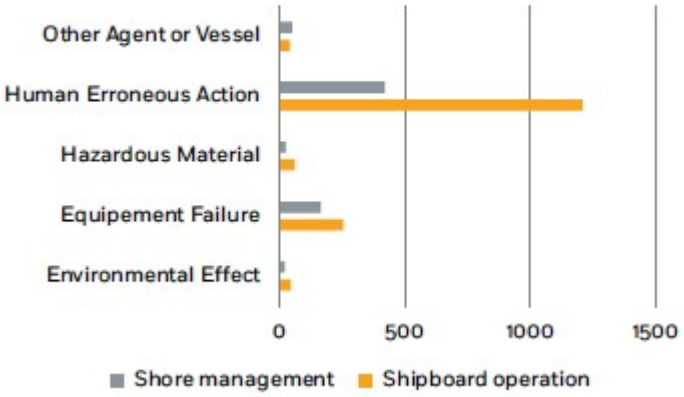
Safety and Shipping Review 2017 (Allianz Global Corporate & Specialty)

All causes of loss

	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	Grand Total
Foundered (sunk, submerged)	69	73	61	64	45	55	70	50	65	46	**598**
Wrecked/stranded (grounded)	35	34	23	24	29	26	21	18	19	15	**244**
Fire/explosion	18	16	14	12	9	13	15	6	7	8	**118**
Collision (involving vessels)	17	13	13	10	3	5	2	2	6	1	**72**
Machinery damage/failure	14	8	7	4	6	15	2	5	2	8	**71**
Hull damage (holed, cracks, etc.)	11	4	8	4	3	7	1	4	2	4	**48**
Miscellaneous	3	1	2	6	1	1	1	2		1	**18**
Contact (e.g harbor wall)	2	1	1			2		1			**7**
Missing/overdue	1	1		1						2	**5**
Piracy	1		1	2	1						**5**
Grand Total	**171**	**151**	**130**	**127**	**97**	**124**	**112**	**88**	**101**	**85**	**1,186**

Source: Lloyd's List Intelligence Casualty Statistics. Data Analysis & Graphic: Allianz Global Corporate & Specialty

Figure 26: Relationship between accidental events and the main contributing factors



Autonomous ships- real and futuristic

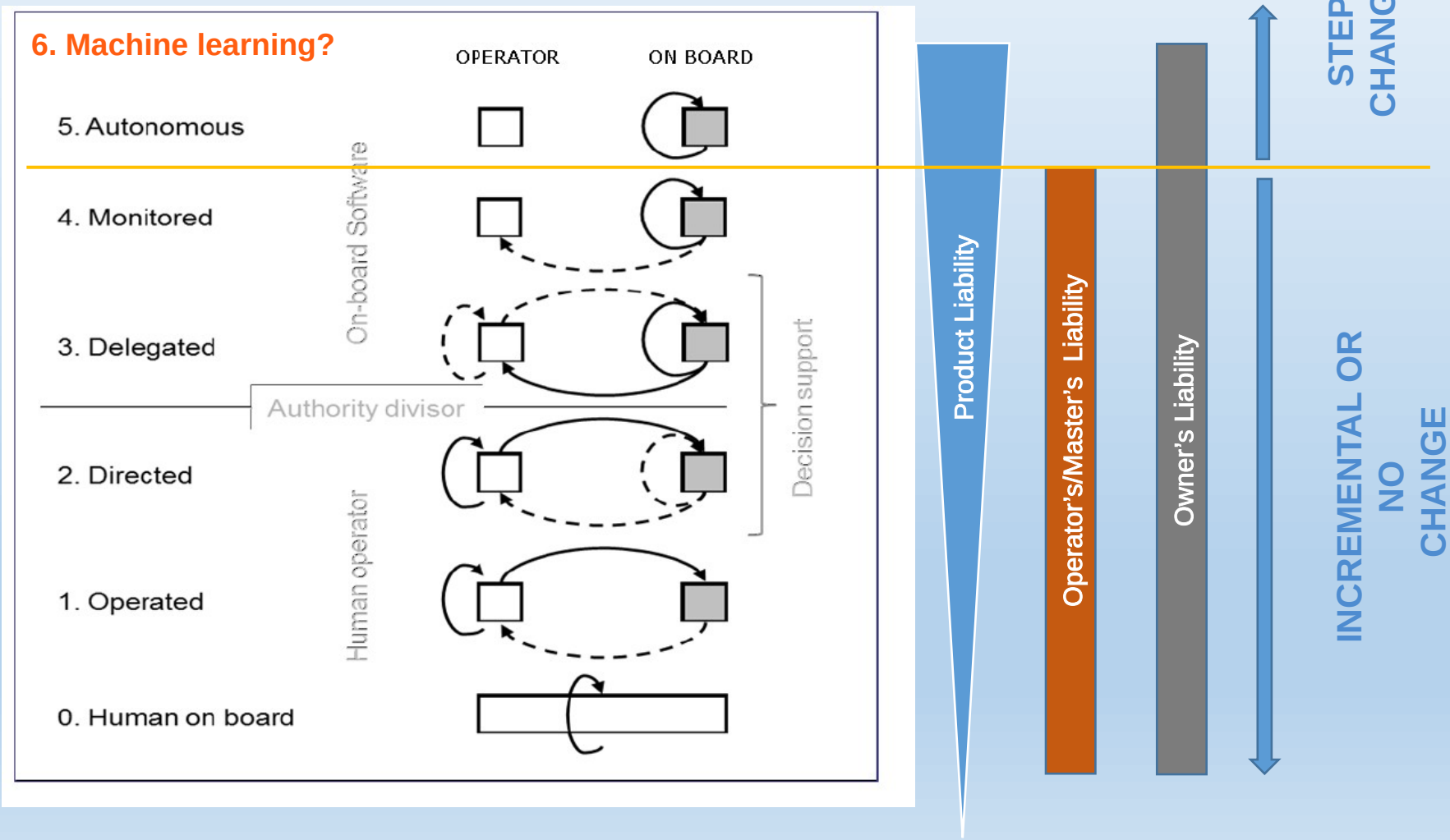
The Sea Hunter, here on a trial outing on the Willamette River in Portland, runs with a crew at the moment but if all works out, it won't need one. (Defense Advanced Research Projects Agency)

An autonomous minesweeper system that can safely clear sea lanes of mines has been handed over to the Royal Navy, Defence Minister Guto Bebb has announced. Crown copyright.

Terminology

- Autonomous : Having its own laws
- Autonomous does not mean unmanned (un-crewed)
- Unmanned does not mean without direct human control (e.g. remotely controlled)
- Autonomous and Unmanned do not mean Intelligent

Level of autonomy and disruption



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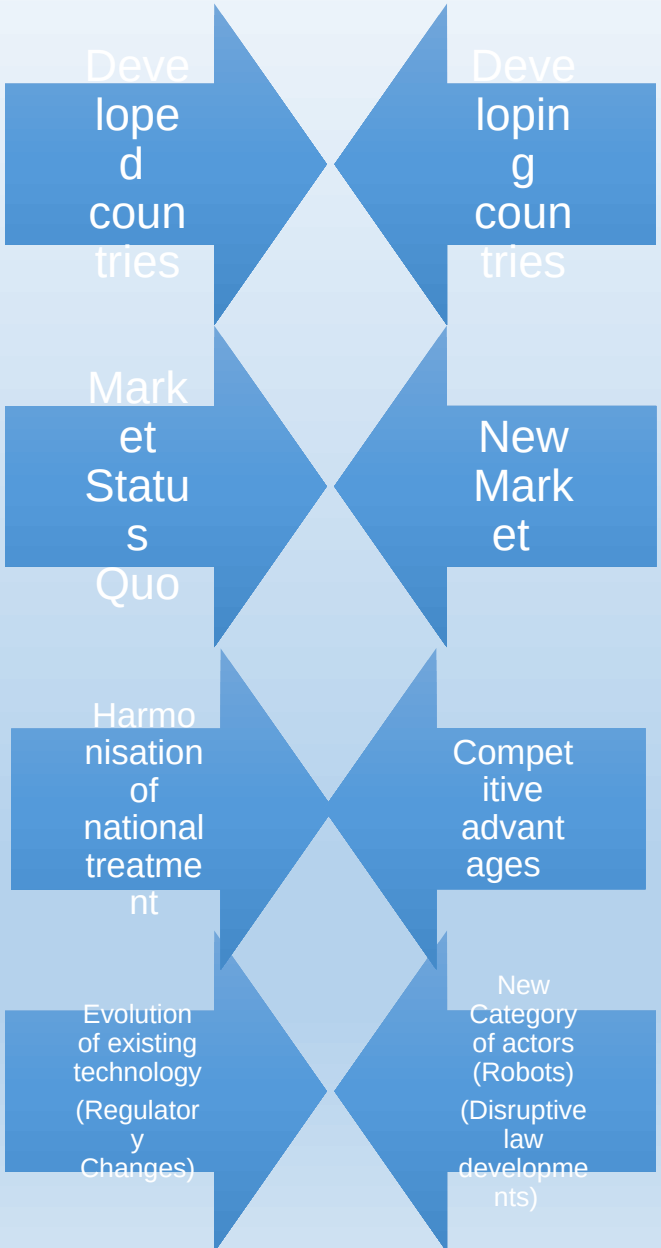
What do YOU think are the regulatory, legal, ethical issues with autonomous and/or uncrewed shipping?

Developing conflicts



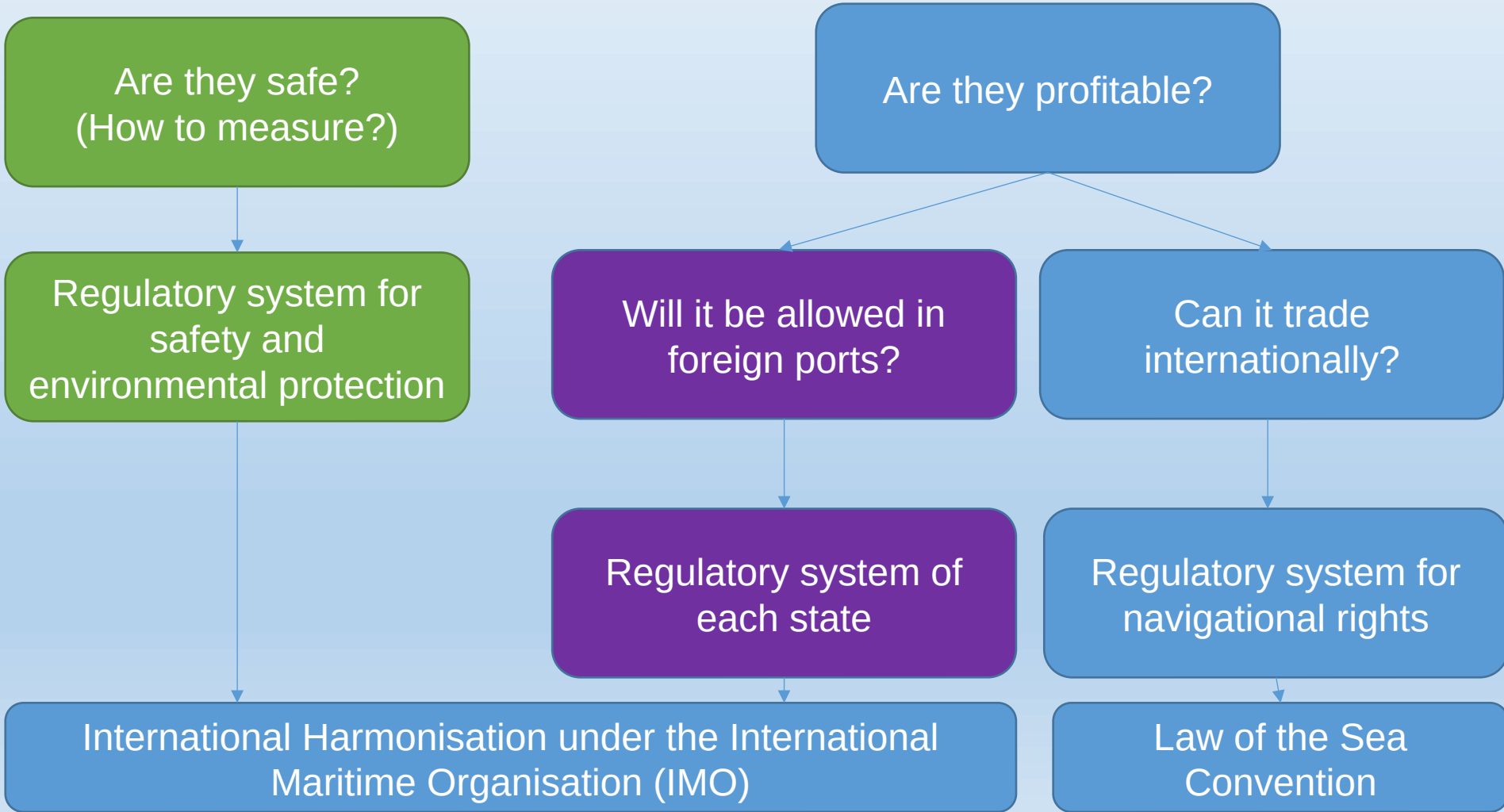
Control of shipping/trade
Economic development
Ownership and business model
Naval Supremacy
Crew employment

Character of shipowning companies
Safety of Navigation
Training and employment

Investment needs competitive advantages
Harmonisation of laws leads to harmonisation of technology

Path of least resistance
Standards for replacing human decision making (civil and naval)
Ethics and decision making

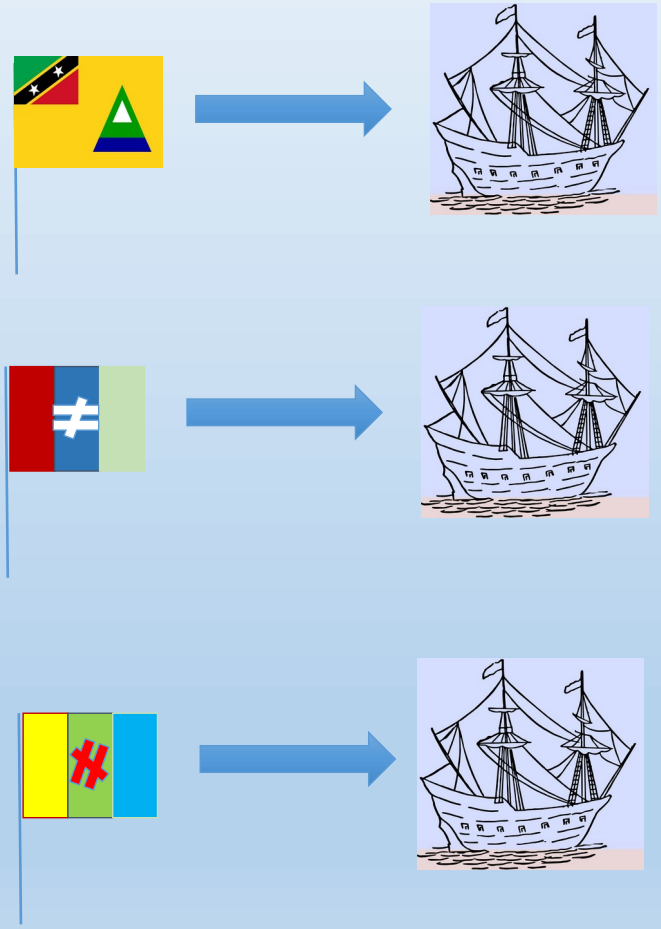
Requirements for success for new types of ships



The Regulatory system for conventional ships

States:
- impose standards in their ports
- agree standards for
  - Construction
  - Design
  - Equipment and
  - Manning
  - And other standards

-issue certificate as evidence for compliance
- Accept CDEM as enough for sailing through the territorial sea
- Accept certificates as prima facie evidence of compliance



Registered Ships:
- Freedom of navigation
- Innocent Passage
- Transit Passage
- Archipelagic passage

BUT ONLY WHEN ASSOCIATED WITH A STATE (see INTERTANKO and Others ECJ C-308/06)

nternational Law National Law

Underlying Legal Norms

- State supremacy- non-negotiable/basis of international law
- Freedom of navigation (various zones)

- Safety of life- known for conventional ships
- Safety of the environment- known for conventional ships

- Facilitation of trade

Regulatory Framework Options

Is an autonomous ship a ship?

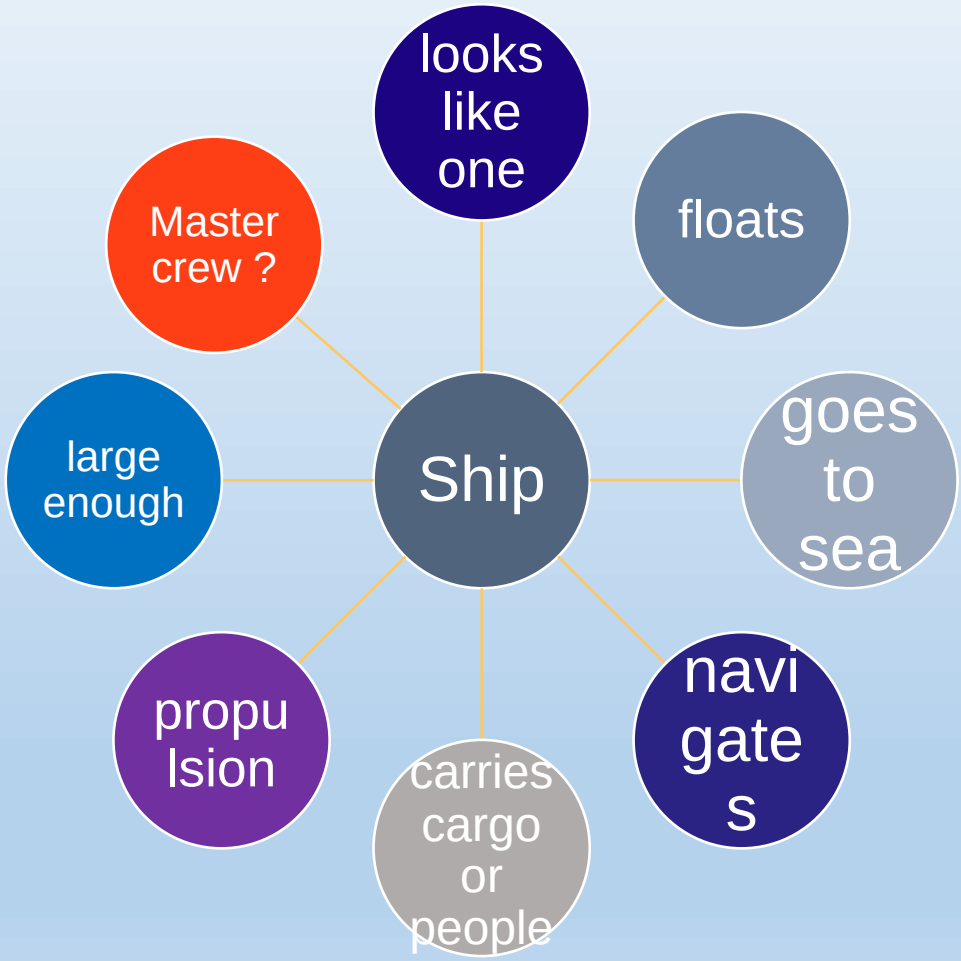
Yes

-IMO Regulations
-Navigational Safety
-Freedom of Navigation
-Limitation rights
-Innocent/Transit passage

No

-National regulatory and safety systems
-Freedom to use at High Seas possible interference
-Innocent/transit passage in doubt

When is something a ship? National laws vary



None is conclusive

The context the question is asked matters

What is a ship?

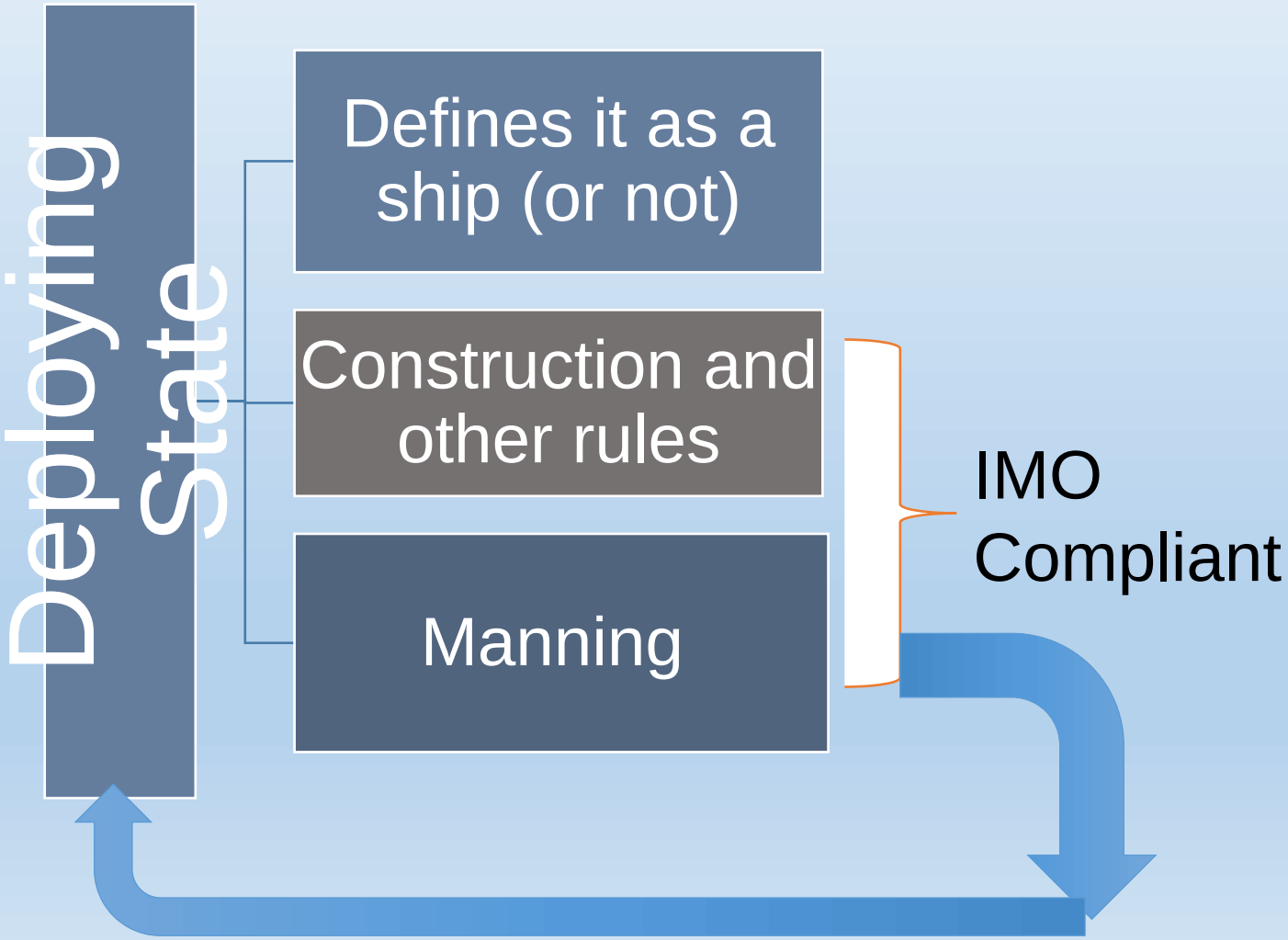
- No internationally accepted definition of ship
- Definitions between national laws differ
  - Under different national laws the same craft may be a ship in one country and not be a ship in another country
- Each national law employs more than one definition depending on the policy objective
  - Under the same national law some autonomous ships will be “ships” or “vessels” some will not be

- Note: the difference in national definitions did not stop the development of shipping,

Legal rights for such craft depend on the operation

- Transport
- Surveying of structures
- Research
- Naval applications
  - Peacetime or wartime
  - Surveillance, minesweeping, transport, hydrography
  - Weapon deployment (Lethal Autonomous Robots)

So who determines the craft's nature?



Do other states have to accept such a definition?

Do coastal states have any powers?

- Assume a state has flagged an unmanned structure as a ship
- What are the options for the coastal state?

Jurisdictional zones

- Ports and internal water
  - Exclusive jurisdiction of the coastal state
- Territorial sea (12 nm)
  - Sovereignty of coastal state
  - “Innocent passage” of foreign **ship**
- Exclusive Economic Zone (200 nm)
  - Exclusive rights with respect to marine resources and the marine environment
  - Freedom of navigation for foreign **ships**
- Areas beyond national jurisdiction (High Seas)
  - All states have rights
  - No state can control or interfere with a foreign **ship** (few exceptions)

But according to whose definition?

So how can the international system work?

- Scenario A
  - If the flag state A considers an autonomous craft as ship
  - The coastal state B through which the crafts navigates (innocent passage also considers it as a ship
  - The port state C at which port the craft is destined also considers it as a ship
- Scenario B
  - If the flag state A considers an autonomous craft as ship
  - The coastal state B through which the crafts navigates also considers it as a ship
  - The port state C at which port the craft is destined does not consider it as a ship
- Scenario C
  - If the flag state A considers an autonomous craft as ship
  - The coastal state B through which the crafts navigates does not consider it as a ship
  - The port state C at which port the craft is destined considers it as a ship
- Scenario D
  - If the flag state A considers an autonomous craft as ship
  - A naval ship of state B, which does not consider the craft a ship, meets the ships on the high seas and considers boarding it

Navigational rights

- It is clear:
  - The deploying state may declare a craft as a ship
  - This will grant the autonomous ship all the navigational rights
  - Only difficulty appears to be when visiting foreign ports which may decide to exclude or detain the ship

Safety of Navigation

- Engineers are obsessed with coding the collision regulations
  - They should be coding “good seamanship”

- Duty of care owed to other users of the sea
  - Safety is the objective
  - Good seamanship is the standard
    - Collision Regulations (vessels)
      - “nothing in [the Rules] shall exonerate any vessel … from the consequences of any neglect to comply with … any precaution which may be required by the ordinary practice of seaman…”
- How safe should autonomous ships be?
  - How could this be demonstrated?

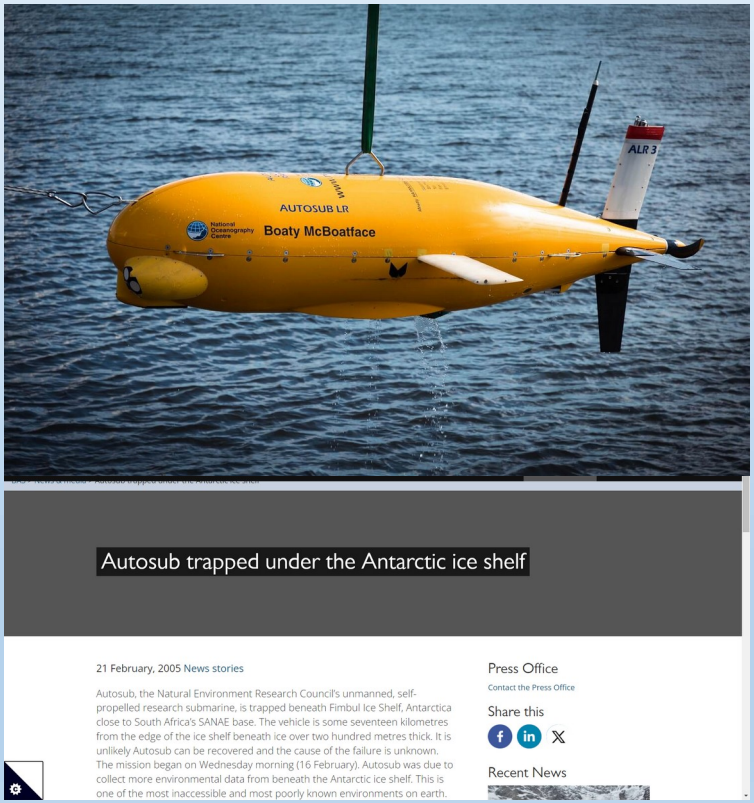
- Consequences
  - Training of operators
  - Good Practice (Seamanship) Guides
  - Risk assessment and management

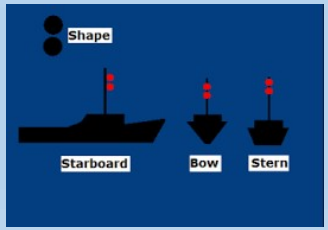
Who sets the safety standards?

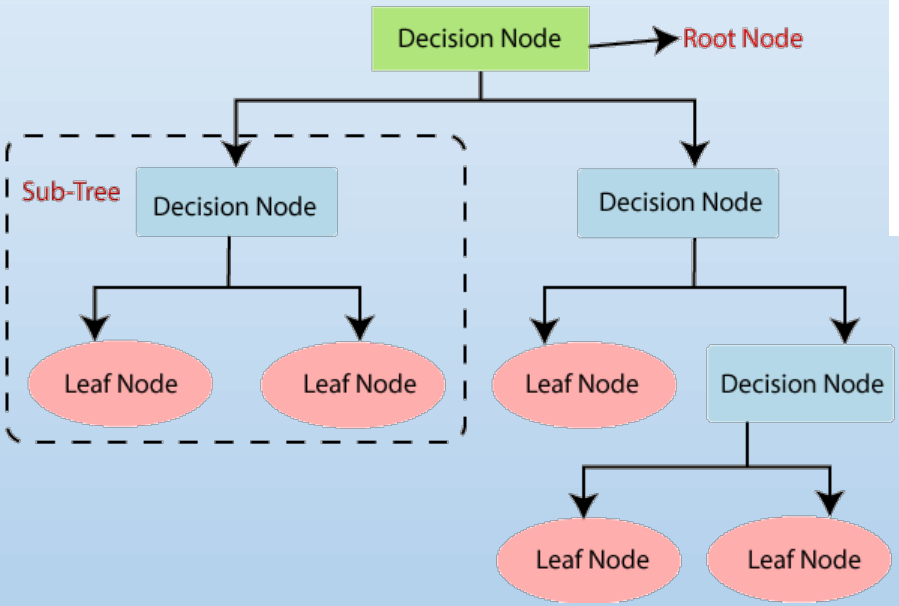
- IMO standards are agreed for all ships
- Enforced by each flag state
- Flag state provides certificates
- Ships are not controlled for safety- the certificates are checked
- If this system is followed for autonomous ships
  - Definitions for autonomous ships should be discussed, accepted by all states and implemented
    - This takes time
    - No innovator can make money- all becomes available to all at the same time
- Harmonisation and standards prohibit investment
  - Industry remains cheap and dirty
  - Ready made solutions can only be implemented
  - GHG emissions - exampls

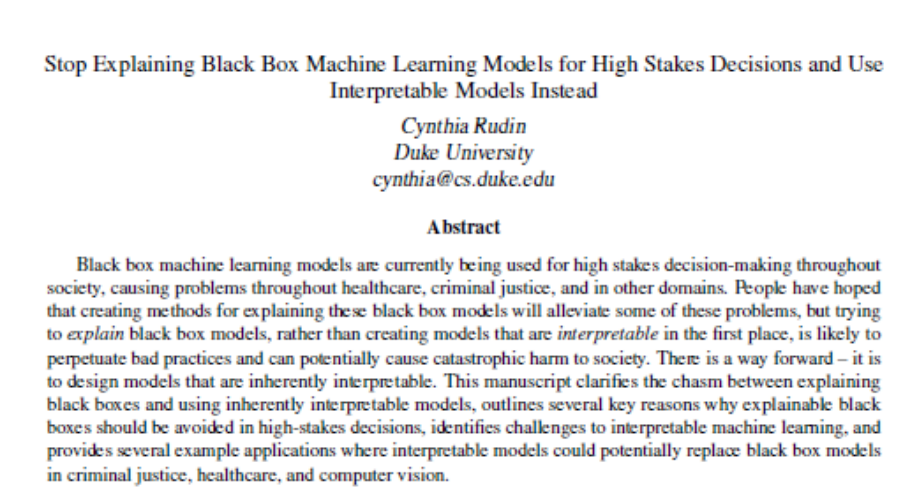
How do we apply the legal, ethical and policy principles?

Recognising targets: Levels of complexity









Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead

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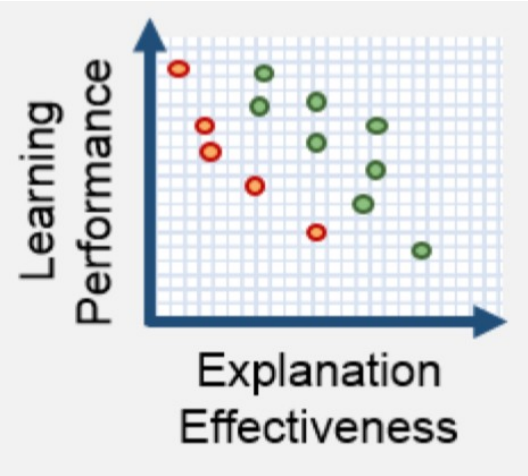
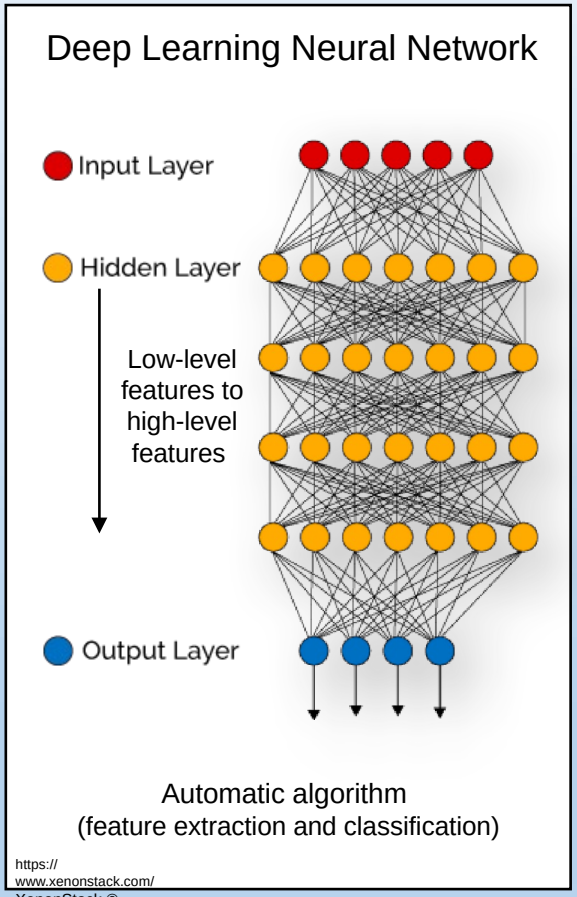
Abstract

Black box machine learning models are currently being used for high stakes decision-making throughout society, causing problems throughout healthcare, criminal justice, and in other domains. People have hoped that creating methods for explaining these black box models will alleviate some of these problems, but trying to *explain* black box models, rather than creating models that are *interpretable* in the first place, is likely to perpetuate bad practices and can potentially cause catastrophic harm to society. There is a way forward – it is to design models that are inherently interpretable. This manuscript clarifies the chasm between explaining black boxes and using inherently interpretable models, outlines several key reasons why explainable black boxes should be avoided in high-stakes decisions, identifies challenges to interpretable machine learning, and provides several example applications where interpretable models could potentially replace black box models in criminal justice, healthcare, and computer vision.



Deep Learning Neural Networks
Architecture and How They Work



https://www.xenonstack.com/
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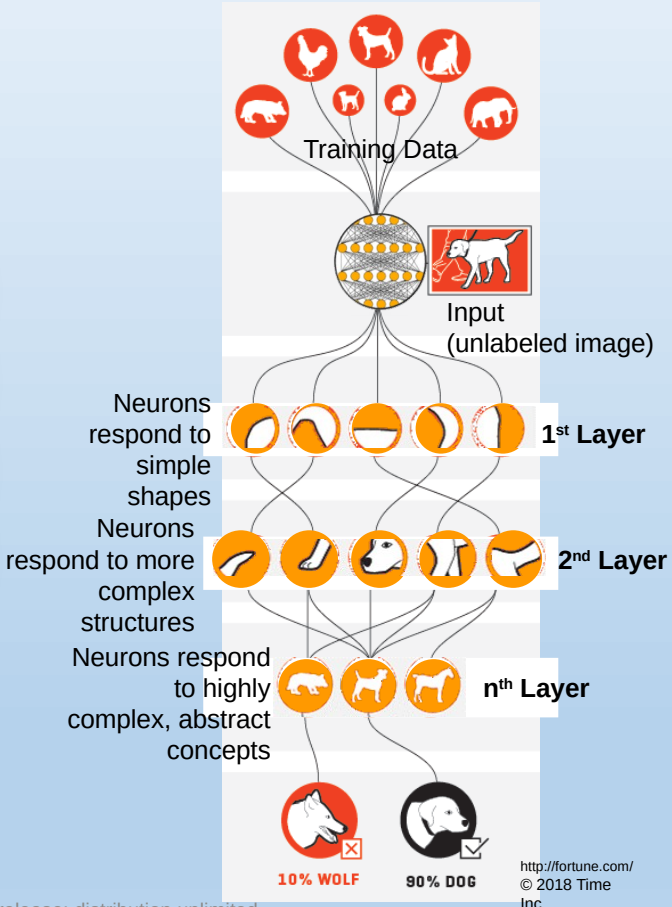

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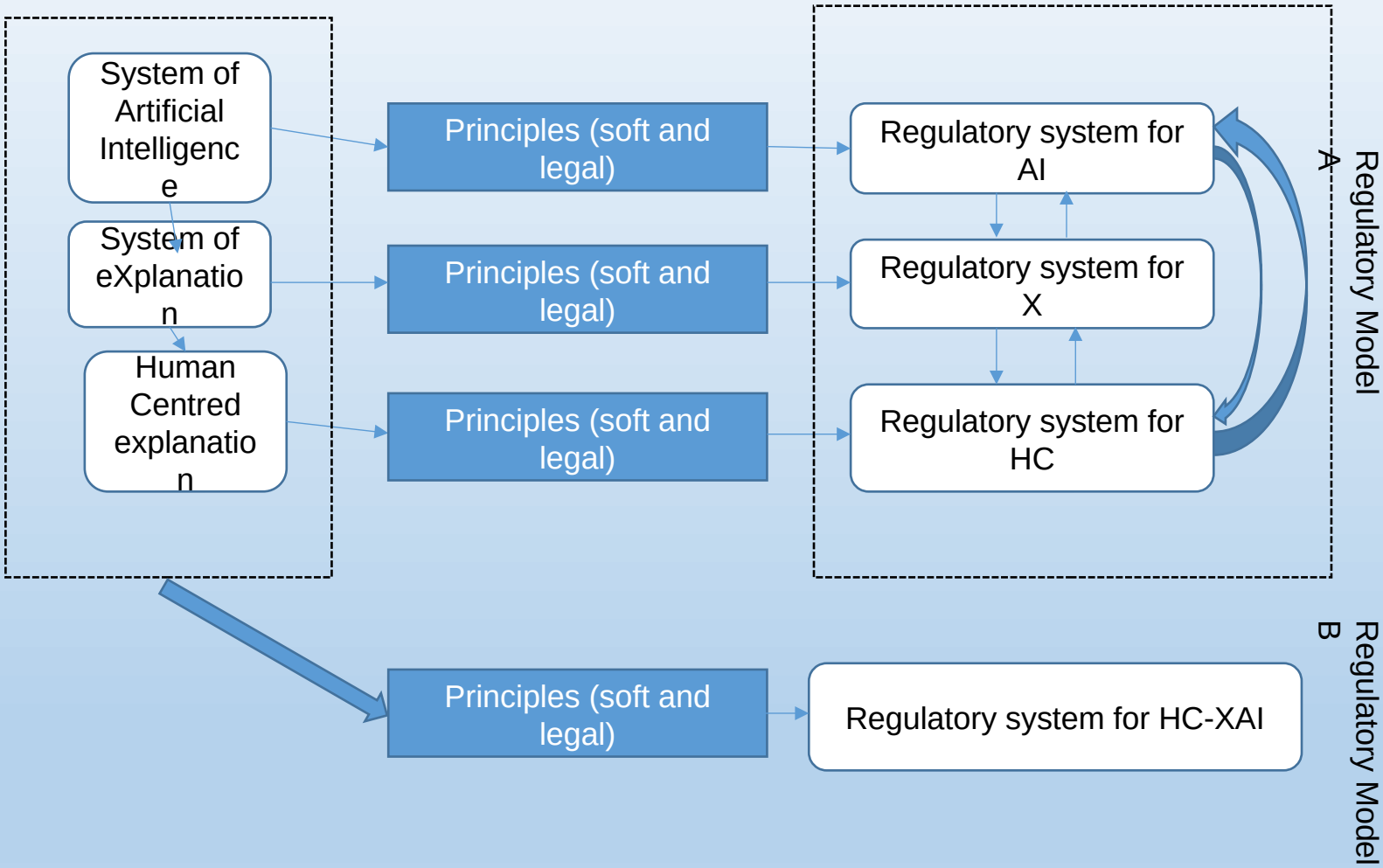
System of Artificial Intelligence
System of eXplanation
Human Centred explanation
Principles (soft and legal)
Principles (soft and legal)
Principles (soft and legal)
Regulatory system for AI
Regulatory system for X
Regulatory system for HC
Regulatory Model A
Principles (soft and legal)
Regulatory system for HC-XAI
Regulatory Model B

Autonomous warships

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"warship" means a ship belonging to the armed forces of a State bearing the external marks distinguishing such ships of its nationality, under the command of an officer duly commissioned by the government of the State and whose name appears in the appropriate service list or its equivalent, and manned by a crew which is under regular armed forces discipline.

Law of the Sea Convention Art.29

Legal status of naval unmanned craft

peacetime

- *If a warship*
  - Entitled to *sovereign immunity*
- If **not a** *warship then:*
  - If controlled by a warship probably shares its legal status
- If fully autonomous
  - Not under “the command”- not a warship
- BUT if exclusively used by a state for government service then entitled to *sovereign immunity* irrespective of characterisation

At times of War

- Naval operations are governed by:
  - customary rules and principles of international law
  - treaties
- Codification of the law of naval operations is unrealistic
  - Codes by navies evidence customary practice – inconsistent
- Apparent conflicts with UNCLOS
  - Self-defence (Law of war) prevails

Capabilities and duties of unmanned naval craft

Intelligence, Surveillance, and Reconnaissance	Targeting	Attacks to legitimate military objectives	✔
Prize measures	Visit	Search and Capture	✕
Identification of targets • Particularly ships of neutral states assisting the enemy	Search and Rescue after engagements • Article 18 of Geneva Convention II	Place passengers, crew and ship's papers in a place of safety before a captured merchant vessel is destroyed as prize. • 1936 London Protocol- for UUVs	✕
Respect neutrality	Avoid indiscriminate attacks	Obligation to take all feasible precautionary measures to avoid civilian casualties	?

Are autonomous systems better in complying with the Laws of War?

- They do not need to protect themselves in cases of low certainty of target identification.
- Self-sacrificing manner can be utilised
- Ability to use an increasing range of sensors better equipped than humans
- No emotions to cloud judgment
- Integrate more information from more sources faster before responding
- Potential capability of independently and objectively monitoring ethical behavior in the battlefield by all parties and reporting violations

Concerns

- Automated systems will not always do only what they are programmed to do
  - Codes too complex and fragmented
  - Machine learning
- What happens when they malfunction?
  - For civilian craft there is a functional framework
  - For naval craft who would be the war criminal?
    - Especially if they are autonomous

To what extent are these real concerns?
Are they really protective regimes for the current technology?

Issues to consider

- Are the armed craft capable of proportional response?
- Does the removal of the cost of human life make conflict less costly and more common?
- Does the perception about relative benefits over humans matter?
- Is the offensive capability appropriate when search and rescue measures not available?
- How contingency planning, pollution incidents etc. can be mitigated?

Conclusions

- The use of unmanned ships for naval purposes is largely covered by existing laws
- Their operational use in peace time is largely unproblematic with general international and national laws filling in ay existing gaps
- Significant ethical and legal problems exist when unmanned ships are used as platforms for automated weapons despite their better capabilities when compared to human operators.
- Regulations depend on the USE of the technology too- and the principles change depending on that use!

- **Questions**
- Would you suggest that these should be developed?
- What are the ethics and moral values for war?
- How could the regulatory system improve to facilitate their use?

Thank you